

VersaTouch: A Versatile Plug-and-Play System that Enables Touch Interactions on Everyday Passive Surfaces

Yilei Shi, Haimo Zhang, Jiashuo Cao, Suranga Nanayakkara
The University of Auckland
Auckland, New Zealand
{yilei,haimo,evan,suranga}@ahlab.org

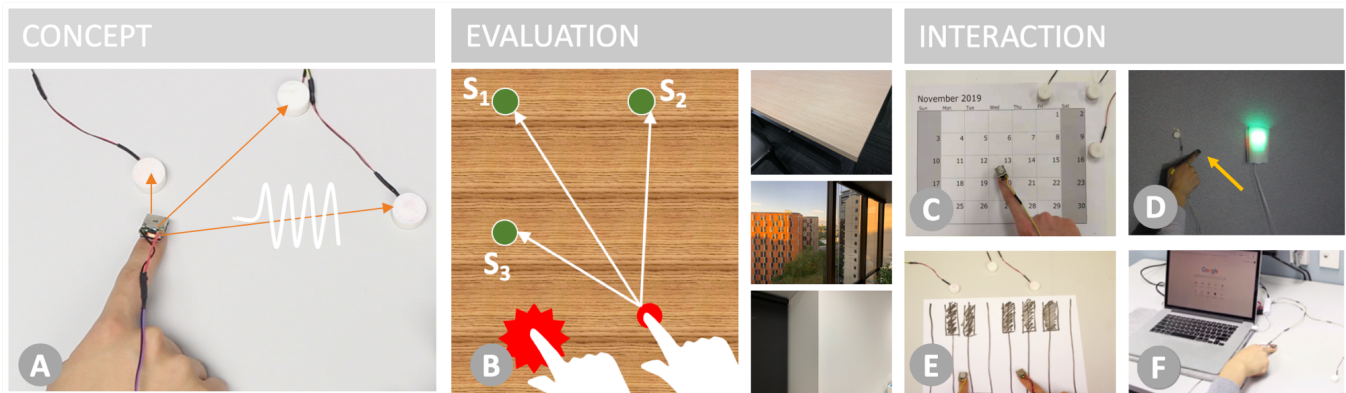


Figure 1: VersaTouch enables various touch interactions on everyday surfaces. A: Active acoustic sensing method was used to localise touch and sense the force. B: Touch localisation and force sensing performance was evaluated on various everyday surfaces. A few examples of VersaTouch interactions C: Checking schedule with specific sensor layout. D: Single sensor as a light controller; E: Multi-touch sensing on hand-drawn pianos; and F: 2-sensor scale-adjustable slider.

ABSTRACT

We present VersaTouch, a portable, plug-and-play system that uses active acoustic sensing to track fine-grained touch locations as well as touch force of multiple fingers on everyday surfaces without having to permanently instrument them or do extensive calibration. Our system is versatile in multiple aspects. First, with simple calibration, VersaTouch can be arranged in arbitrary layouts in order to fit into crowded surfaces while retaining its accuracy. Second, various modalities of touch input, such as distance and position, can be supported depending on the number of sensors used to suit the interaction scenario. Third, VersaTouch can sense multi-finger touch, touch force, as well as identify the touch source. Last, VersaTouch is capable of providing vibrotactile feedback to fingertips through the same actuators used for touch sensing. We conducted a series of studies and demonstrated that VersaTouch was able to track finger touch using various layouts with average error from 9.62mm to 14.25mm on different surfaces within a circular area of 400mm diameter centred around the sensors, as well as detect

touch force. Finally, we discuss the interaction design space and interaction techniques enabled by VersaTouch.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**; *Ubiquitous and mobile computing systems and tools*.

KEYWORDS

Finger Touch, Localization, Force, Acoustic

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1 INTRODUCTION

Our fingers are incredibly agile and expressive, and we constantly feel the natural haptic feedback of objects while interacting with them. Today, touch interactions on commercialised products are generally supported by touchscreens and touchpads. Despite their accuracy in touch tracking, these specially manufactured devices remove natural haptic feedback, as well as confine touch sensing capability to limited areas.

Various technologies have been developed to sense finger touches on ordinary surfaces. A popular method is to use computer vision [9, 17, 23, 26, 45, 47]. This method requires the line of sight, and is sensitive to external factors, such as lighting. Alternatively,

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electric field tomography [51, 53] and capacitive sensing [37, 39] could also be embedded in the surfaces to track touch. However, these techniques usually involve a complex setup or instrumentation of the surface, limiting the portability of the sensing system. Alternatively, acoustic sensing technology has been used to detect touch. For example, passive acoustic systems leveraged the sound of discrete knocking to track its position [34, 35] or direction relative to a device [48]. Active acoustic sensing has also been used to detect discrete finger touch [2]. There has been also a lack of previous work that are able to detect force, the location of a touch and provide vibrotactile feedback in a single system.

In this paper, we present VersaTouch, a portable and versatile system that enables two distinctive input modalities of touch, location and force, on everyday surfaces, without permanently instrumenting the surface. The system also supports vibrotactile feedback on the fingertips, as a “free” additional affordance by the sensing paradigm. Our system uses an active acoustic signal sensing paradigm, with a fingernail-worn signal source and multiple standalone sensors. The system localises touch using the Time of Arrival (ToA) [7] method, senses the touch force as well as two speeds of landing and release of the touch (hereafter referred to as “dynamic tap gestures”) by analysing the signal intensity as a result of the change of finger deformation. Multiple fingers can be tracked by time-multiplexing the actuation of multiple signal sources. Additionally, vibrotactile feedback is provided by emitting a low-frequency burst from the same signal source. To start using the system, one can simply place the sensors on a surface and attach the signal source to the fingernail. Simple calibration for touch localisation is needed when the layout of the system changes, with only three touches. Our system is versatile along multiple dimensions: (1) *Sensing ability* - as it supports two modalities of touch input, location and force sensing on multiple fingers; (2) *Natural feedback* - as it allows users to feel the surface textures, as well as provide additional vibrotactile feedback on top of the fingertip; (3) *Configuration of the system* - as users can put three sensors in different shapes and use the system after simple calibration; (4) *Feasible surfaces* - as it can be used on various rigid surfaces; (5) *Interaction possibilities* - number or layout of the sensors could be changed according to different use cases. For example, using three sensors, the system can locate the 2D position and transient force of finger touch on a surface; with just a single sensor, the finger-to-sensor distance or dynamic tap gestures could be tracked to control a 1D parameter; and with multiple signal sources, the system can track multiple fingers of a single user or multiple users. We contribute with:

- Design and development of a versatile, plug-and-play active acoustic sensing system, VersaTouch, that can sense multi-finger touch locations and touch force on regular rigid surfaces.
- Empirical evaluations of VersaTouch capabilities including the accuracy and stability of touch tracking and force sensing on various everyday surfaces.
- Exploration of the interaction design space and interaction techniques that are enabled by VersaTouch.

2 RELATED WORK

In the following section, we review three types of related work: touch interactions on everyday surfaces, acoustic based localization

and touch pressure sensing. In Table 1, we compare VersaTouch with systems that sense finger touch on everyday surfaces.

2.1 Touch Interaction on everyday Surfaces

Many methods have been put forward to enable touch interactions on everyday surfaces. A simple solution is to instrument the surfaces with various sensors. SmartSkin [37] lays a mesh of transmitter/receiver electrodes on a table and tracks the hand with capacitive sensing. Similarly, Electrick [52] and Pulp Nonfiction [51] coat everyday surfaces and papers with a conductive layer to track touch locations using electric field tomography. Although there are a few inherently conductive objects, such as our skin [54], most easy-to-access everyday surfaces still need the augmentation of conductive layers, resulting in extra costs and effort. Computer vision based technologies were able to avoid instrumentation of objects. I/O Bulb [40] used a pair of cameras and projectors to create interfaces on normal desks. Recently, depth cameras have been adopted to sense touch on various everyday surfaces [15, 45–47]. Thermal cameras [17] was also able to sense touch in real-time. Although instrumentation of surfaces is not necessary, camera-based sensing technologies are influenced by occlusion and lighting conditions [37]. Besides, extra sensing methods [8] were also developed to improve the touch sensing accuracy. Finally, acoustic sensing was also used to sense touch on everyday surfaces [14, 35]. Acoustic sensing technology has the potential to be designed as a portable unit, such as Toffee [48]. However, Toffee can only sense discrete knocks from different directions and is unable to localize the exact positions. There is still a gap of exploring portable systems for touch interaction with acoustic sensing technologies.

2.2 Acoustic Based Localization

2.2.1 Passive Acoustic Localization. Passive acoustic localization systems, which do not emit acoustic signals themselves, have been widely applied in various situations. PingPongPlus [14] tracked the landing point of a ball with eight microphones mounted under a table. The actual location was calculated using the Time Difference of Arrival (TDOA). TDOA has also been applied to localize knocks on glasses [34, 35] or other tangible objects [36]. Toffee [48] attached four piezoelectric sensors to the backside of devices to enable around-device knocking interactions. Skinput [10] localized the finger taps on human body by modelling the propagation of mechanical vibration along the skin. Inertial sensors of smart-watches have been leveraged to localize finger taps on the backside of hands [16, 50]. Above research mainly focused on localizing discrete tap events and were unable to sense continuous touch movement. By searching the maximum in the likelihood map, Yang et al. [49] was able to track touch trajectory relying on the sound generated by finger movement. However, it is still difficult to perform some basic touch interactions, such as “touch” and “hold”. A real example could be dragging and holding a file on a folder to open its sub-folders on Mac. Since their algorithms interpret “holding” gestures as “no interaction”, the system would divide the whole interaction into two. It is an intrinsic limitation of passive acoustic sensing that the sensing capability highly relies on the sound generated by the movement. Besides, these techniques are unable to sense multi-touch interactions due to the interference from different sound sources.

Table 1: Summary of prior work on touch tracking technologies.

	Technique	Touch Tracking	Dynamic Force	No Surface Alteration	No Finger Instrumentation	Feedback	Working Surface	Resolution
Electrick [48]	Electrical	Continuous touch	No	No	Yes	No	Any Surface	0.9~2.7cm
Skintrack [50]	Electrical	Continuous touch	No	Yes	No	No	Skin of Arm	0.7~2.4cm
Pulp Nonfiction [47]	Electrical	Continuous touch	Yes	No	Yes	No	Paper	~1.4cm
DIRECT [42]	Computer Vision	Continuous touch	No	Yes	Yes	No	Any Surface	~0.5cm
Tangible Acoustic [32]	Passive Acoustic	Discrete tap	No	Yes	Yes	No	Rigid Surface	3~5cm
Toffee [44]	Passive Acoustic	Angular tap	No	Yes	Yes	No	Rigid Surface	Non-localisation
Continuous Acoustic Source Tracking[45]	Passive Acoustic	Continuous touch	No	Yes	Yes	No	Rigid Surface	~1.0cm
Sound of Touch [23]	Active Acoustic	Discrete tap	Yes	Yes	No	No	Skin of Arm	Non-localisation
Touch&Activate[27,28]	Active Acoustic	Discrete tap	Yes	Yes	Yes	No	Rigid Objects	Non-localisation
VersaTouch	Active Acoustic	Continuous touch	Yes	Yes	No	Yes	Rigid Surface	~1.0cm

2.2.2 Active Acoustic Localization. An early work of active acoustic localisation was the sonic pen [1]. It generated short sonic pulses, which were captured by an orthogonal pair of plane microphones to get the position of the pen. Touch & Activate [32] attached a contact mic and a sound generator to everyday objects and used machine learning to classify discrete finger touches. Swadloon [13] realised indoor localisation by measuring the phase and frequency shift of acoustic signals from a phone. By using distributed Frequency Modulated Continuous Waveforms (FMCW) and Doppler shifts, CAT [19] estimated the relative distance and velocity of a smartphone that was emitting designed signals. The distance and velocity were then fused to track the movement of the phone. Research that were more related to our work are techniques tracking finger touches with active acoustic sensing. For example, an active acoustic touch interface [2] was developed using a single transmitter with no more than two receivers. It localized finger touch positions by analyzing the changes in the frequency response of the surface due to the touch events. Other techniques, such as Low-Latency Acoustic Phase (LLAP) [42] and Orthogonal Frequency Division Multiplexing (OFDM) [30], leveraged the existing microphones and speakers of smart devices to enable around-device finger tracking interactions. Recently, active acoustic sensing was used to detect touch in projected touch interactive system, such as Touchable Wall [12]. Although the capability of touch localisation and force sensing were introduced, no evaluation was conducted and the localisation was coarse. The work most related to ours, is the Sound of Touch [27], which was based on transdermal low-frequency ultrasound propagation. It enabled press-aware continuous touch sensing, as well as gesture detection. However, this still differs from our work in the following two aspects. First, it mainly focused on touch interactions on arms, which are cylindrical and deformable. Little was investigated whether this technique worked for flat and rigid surfaces. Second, Sound of Touch looked into detection of discrete touch positions and binary force touch. Our work, on the other hand, focuses on fine-grained touch localisation and multiple pressure level detection.

2.3 Touch Pressure Sensing

2.3.1 Instrumenting the Surface. The force of touch has been leveraged as an extra modality to augment interactions [3, 4, 22, 25].

A widely-used technique to detect force is to use Force Sensitive Resistors (FSRs), which is a polymer film and changes resistance according to the applied force. For instance, GraspZoom [24] attached FSR to the backside of a smartphone and enabled one-handed zooming and scrolling control. A similar method was also used to detect force press at the edge of a smartphone [44] and force gestures performed when holding an iPod [11]. UnMousePad [38] created a flexible and inexpensive multi-touch device with Interpolating Force Sensitive Resistor (IFSR), which can sense three levels of pressure. Other technologies were also applied to capture force. GelForce [41, 41] realized force detection on a transparent elastic surface. The force was detected by annotating the surface with colored markers and analyzing the visual deformation of these markers. Dietz et al. [5] embedded piezoresistive materials into the membrane of a physical keyboard, making the whole keyboard pressure sensitive. The above methods detect force or pressure by permanently augmenting an interactive surface, which is usually accurate but not easy to remove or move to another surface. Finally, Ono et al. [33] used active acoustic sensing to train machine learning models to detect the force touch on discrete locations. The system first used a support vector classification (SVC) model to predict the touch location. Then, it recognized the force level with a corresponding support vector regression (SVR) model. One limitation of the system was that it requires users to retrain the SVC model and train a new SVR model for a new touch location.

2.3.2 Instrumenting the Finger. Other force detection techniques were also developed without permanently augmenting the surface. An armband was designed to sense one dimensional tangential force with photo-reflective sensors [18], which measures the gap between the sensor and the skin as an estimate of the tangential force. Low-frequency ultrasound [27] has also been used to sense on-body touch gestures, as well as binary levels of touch force. In addition to an armband, various finger wearable devices were designed also to enable force touch. For example, when a finger presses against a surface, the alteration of the blood pattern underneath the fingernail could be leveraged to predict various kinds of force and finger postures [20]. Another fingernail wearable system [28] sensed the fingertip deformation with laser displacement sensors. These two systems are relevant to VersaTouch, since VersaTouch is based on deformations of fingertips. However, in VersaTouch, we focused on

enabling touch tracking, force detection and vibrotactile feedback in a single system that offers diverse interaction possibilities.

3 VERSATOUCH

VersaTouch system uses active acoustic sensing technique. Compared to passive techniques, active acoustic sensing can sense gentle touch as well as continuous movement [2] due to its stable and known signal pattern. When the finger touches the surface, the sensors capture the acoustic signal emitted by the finger-mounted transducer (Figure 3A). Then, the system predicts the 2D location and the rate of change of the touch force by analyzing the signal.

3.1 Touch Localisation

As we described in the paper, VersaTouch could be used with 1 sensor, 2 sensors or 3 sensors for various scenarios. However, 3 sensors are needed to support 2D continuous touch localisation. In this section, we describe the basic theory behind this, i.e., distance measurement of touch to each sensor as well as the methods to localisation. Time of Arrival (ToA) and trilateration [31] are used to localise the touch event. The ToA method measures the travel times of the active acoustic signal from the fingertip to each receiver, which is used to approximate the emitter-receiver distances. The system then uses trilateration to determine the exact touch location.

We used the first clear low trough that is detected by each sensor (Figure 2) as the time of arrival of the signal at that sensor. The subsequent portion of the signal can potentially be contaminated by the echo of the signal itself. From our experiments, the first clear trough demonstrated consistent clarity among most surfaces we tested. The time period from the beginning of the reading to the first clear trough is measured, represented by T_i for respective sensors (Figure 3). The measured time, T , consists of two parts: T_t and T_d where $T = T_t + T_d$. T_t is the exact travel time of the signal from the touch point to the sensor and T_d is the combined system delay from other factors. The main part of T_d is an offset between the first clear trough and the actual arrival time, which should be earlier. Factors contributing to T_d include the travelling time of the signal through the finger and the delay in the sensing system. While T_t is dependent on the touch location, T_d is independent of the touch location. Therefore, it is possible to determine T_d through a calibration process. Although T_d can be dependent on the size of the user's fingertip, in our explorations, T_d was almost constant across all users participating in our user studies.

The finger-to-sensor distance is the product of travel time and speed of sound in the transmission medium. Since the speed of sound is constant in a medium, and different across different materials, we directly use the numerical value of travel time as an approximation of the distance (i.e., up to a constant factor). Particularly, the smallest “distance” measurable in our system is the time interval between two sample points, which is the highest temporal resolution achievable by our system. Another variable of the system configuration is the location of the three sensors, which will be described in the “Calibration” section. With positions of sensors, distances to sensors and time of delay known, we used stochastic gradient descent (SGD) to find the most probable touch location that minimizes the loss function, which is defined as mean square distance error of the predicted touch point:

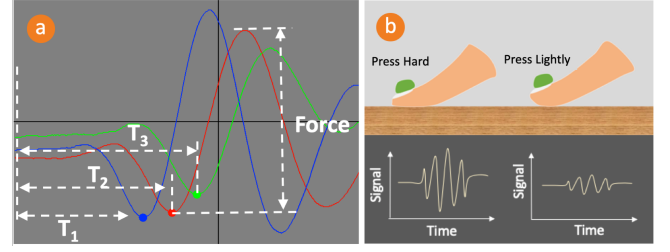


Figure 2: VersaTouch uses the same sensing paradigm to sense touch position and force. a) Red, green and blue curves respectively show the signal received from 3 different sensors; b) Hard press causes a significant squeeze of the finger, which results in a stronger signal emitted on to a surface.

$$Loss = \sum_{i=1}^3 (T_i - \sqrt{(x - x_i)^2 + (y - y_i)^2})^2 \quad (1)$$

where (x_i, y_i) and T_i respectively mean locations of the three sensors and the arrival times they detected. Gradient descent optimization stops when the difference of successive two loss values is less than a predefined threshold.

3.2 Calibration

VersaTouch requires the locations of the three sensors to localize finger touch. We developed a simple calibration procedure to measure the distances between each pair of sensors and accordingly created a coordinate system to localize the touch. Calibration is necessary for different layouts and surfaces. According to our study, the system works well for a walk-up scenario, where a calibrated system could be used by different users with acceptable accuracy.

In most situations, the three sensors would not be collinear and would form a triangle. Since our system tracks the relative touch position with respect to the sensors, we conveniently construct a coordinate system as follows. We set the circumcenter of the triangle as the origin, and direction from sensor 1 to sensor 2 as the positive X direction of the coordinate system (Figure 3B). The calibration then reduces to finding the lengths of three edges of the triangle formed by the sensors. To measure that, the user needs to perform three touches on the table, as shown in Figure 3C. These touches should be outside of the triangle and respectively collinear with each pair of the sensors. The length of each edge can then be calculated as the absolute value of time difference of arrival of the signal captured by two sensors located on the two nodes of that edge. An acceptable calibration result does not strictly require the touch point to be located on the line formed by two sensors. For example, if the calibration touch point is 1cm away from the line formed by two sensors (Figure 3D), the calibrated distance between two sensors was subject to an error of 0.2mm, which is negligible.

3.3 Multi-touch

VersaTouch can support multi-touch interactions. This is realized by attaching multiple surface transducers onto different fingers and time multiplexing their activations. One implication of time multiplexing is that, with more fingers added to the system, the touch tracking sampling rate for each finger would decrease. With

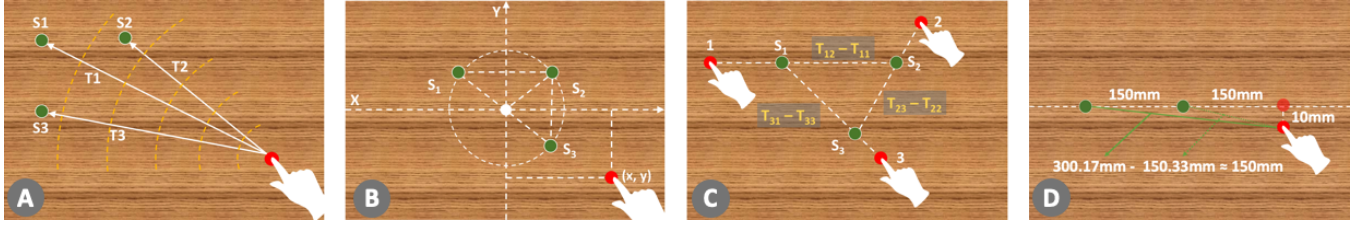


Figure 3: Operation theory of touch tracking. A: Localization with trilateration. B: Coordinate system. C: Touch positions for calibration. D: The displacement of the touch point during calibration would result in a negligible error.

one finger, our system can track touch at a sampling rate of 120Hz. To enable two-finger touch interaction, the sampling frequency has to be reduced to 60 Hz, which did not create a significant perceivable delay.

3.4 Force Touch

Since the signal is emitted by the transducer mounted on the fingernail, the signal would propagate through the finger before it reaches the surface being touched. When a user touches a rigid surface, the fingertip would deform due to the force applied. This force compresses the finger, changes the hemodynamic state and makes the tissue in the fingertip more compact. Empirically, we found that when more force is applied, the received acoustic signal is stronger (Figure 2b). The signal attenuation along the propagation path follows specific attenuation theory¹ $A = A_0 \times e^{-\alpha z}$, where α , x , A and A_0 are attenuation coefficient, distance, signal intensity at measuring location and touch location. With a sample data set, we can collect initial parameters of the formula for each sensor. Since we know the position of each sensor as well as the touch points in the system, we can use the formula to calculate the touch force with respect to each sensor. Finally, we used the average value of the force to represent the actual predicted touch force.

3.5 Feedback

VersaTouch enables two types of feedback: 1) With VersaTouch, users can still receive the natural haptic feedback from the surface, as VersaTouch does not block the mechanoreceptors in the fingertip. 2) Additionally, VersaTouch provide synthetic (i.e., artificial) vibrotactile feedback using the same transducer that acts as the signal source for touch tracking. For this we used a low frequency (500Hz) burst that lasts for 50ms. The intensity and duration of the feedback could also be changed for various interaction scenarios.

4 IMPLEMENTATION

4.1 Hardware Design

We used a small-size surface transducer² to emit the signal. The size of the transducer is 21.5×14.5×7.9mm, suitable for attaching on a typical fingernail. Three 20mm piezoelectric sensors are used to receive the signal. To ensure proper contact between the piezoelectric sensors and the surface, we attached a 3D-printed cylindrical 3-gram weight to each piezoelectric sensor. The size of the

piezoelectric microphone can affect the sensing accuracy. Small piezoelectric sensors produce weaker signals, which result in a low signal-to-noise ratio. This significantly reduces the accuracy of our touch detection algorithm. Large piezoelectric sensors have a larger detecting area, which reduces the spatial resolution. We chose 20mm-diameter piezoelectric, which strikes a balance between signal clarity and spatial resolution. The receivers are connected to an Arduino Due³ with an AT91SAM3X8E micro-controller. The clock frequency of the micro-controller is 84 MHz. A Teensy3.6⁴ was used to control the transducers to generate acoustic signals. One MAX98306⁵ and three MAX9814 amplifiers⁶ amplify the driving signal and the received signal respectively. A 15" MacBook Pro (2.5 GHz Processor, 16 GB 1600 MHz Memory) is used to process the data and calculate the touch positions with a Java program.

4.2 Software Design

Two programs are developed for collecting and processing data respectively. They run separately on the Arduino and the MacBook Pro.

4.2.1 Data Collection: The Arduino program handles signal emission and data collection. Upon reception of a command from the

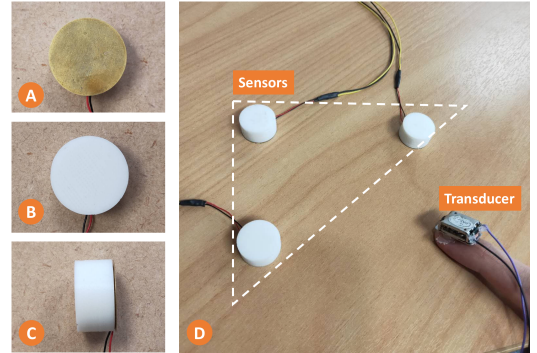


Figure 4: VersaTouch prototype was made with piezoelectric microphones and surface transducer. A, B, and C shows that the sensor consists of a piezoelectric element attached to a weight. (D) The signal is emitted by the transducer and propagates through the table, finally received by the sensors.

¹<https://www.nde-ed.org/EducationResources/CommunityCollege/Ultrasonics/Physics/attenuation.htm>

²<https://www.sparkfun.com/products/10917>

³<https://store.arduino.cc/usa/arduino-due>

⁴<https://www.sparkfun.com/products/14057>

⁵<https://www.adafruit.com/product/987>

⁶<https://www.adafruit.com/product/1713>

computer, the Arduino sends a digital pulse to the Teensy. The latter then generates a 2KHz pulsed sine wave through its DAC ports and drives the surface transducer. Each pulse consists of one wavelength of the sine wave. The system sends acoustic pulses at an interval of 8ms, allowing enough time for the prior acoustic signal to dissipate. At the same time of the pulse generation, the Arduino starts to continuously read 1000 data points from the 3 sensors in 1.78 milliseconds with a sampling rate of 560 KHz. The speed of sound is usually faster in solid material, such as wood, than in air (around 340 m/s)⁷. 1.78ms is long enough for the signal to propagate through a distance of at least 0.6 meters. After all the data is collected, the Arduino transmits the data to the computer through the serial port and waits for the command to trigger the next sensing cycle.

4.2.2 Data Processing: A Java program processes the data. Subsequent to data collection, we smoothed the data with a 20-sample-wide moving average filter, which reduces the high frequency noise and smooths the acoustic signal. Then, we sum up the absolute value of discrete derivative of a moving window of 210 data points. When it reaches a predefined threshold, the signal is assumed to be from a touching finger and the local minimum of the corresponding window is registered as the arrival time for this sensor. For our three-sensor layout experiment, only if all three sensors registered touching action, then the system registers the existing status as “touching”. The touch position is predicted according to the method mentioned in previous section. Significant predictive errors, although rare, may arise due to two reasons: 1) there is a sudden environment noise that reduces the signal-to-noise ratio; 2) the SGD algorithm finds a local optimum instead of global one. These outliers are usually manifested as discrete large jumps from the previous predicted location for a very short period of time, forming a small portion of the data. We use a sliding-window median filter to remove the outliers. Specifically, during testing, we found that a median filter with a window size of 7–11 data points works well for our system. Here, we use a window size of 9 data points. In addition, we used an algorithm to predict the pseudo force of the touch. The algorithm was established based on our empirical study, which will be detailed in section 6.3.

5 STUDY 1: TOUCH LOCALISATION

Objective of this study was to evaluate VersaTouch’s touch localisation accuracy from three aspects. First, we evaluated touch localisation accuracy in terms of four layouts. Then we tested the robustness of touch localisation when the finger is moving. Finally, we evaluated its accuracy on four different materials in real world with normal environment.

5.1 Apparatus and Stimuli

The tested area was a 400mm×400mm square (Figure 5A). We created a coordinate system whose origin was at the centre of the square and the X-axis was parallel to an edge of the square. The tested area covered the coordinate from (-200,-200) to (200,200), where the unit of coordinate was 1mm. To clearly show the coordinate to participants, we marked the space at 100mm intervals,

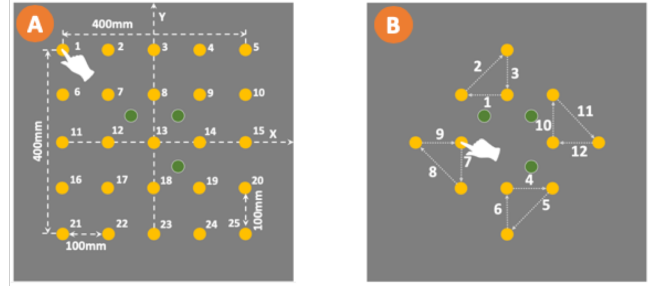


Figure 5: The interactive area evaluated was a 400mm×400mm square. A: Evaluating touch localisation on 25 touch points which were 100mm away from each other; B: 12 touch locations out of 25 were selected to test the continuous touch tracking. The continuous touch was along the edges of the triangles.

forming a 5-by-5 grid (Figure 5A). These points mark positions of the study stimuli, which will be described later. For discrete touch localisation, we tested four different sensor layouts, i.e., three isosceles triangles with their shorter edges of 150mm and an equilateral triangle (Figure 6A) of which each side is 100mm. For all layouts, the origin of the coordinate system is located at the circumcenter of the triangle. We then tested VersaTouch three everyday surfaces made out of different materials: wooden table, glass window, and interior wall made from gypsum board. Finally, we tested VersaTouch’s performance during finger movement. Since we have already investigated the influence of layout, we use only the sensor layout with the higher accuracy, which is a right isosceles triangle with shorter edges of 150mm, for both material and finger movement evaluation.

For the discrete touch localisation, both layout and surface effect evaluation, we tested the localisation accuracy of all 25 touch points. For continuous touch tracking, our aim was to test the viability of our system in a more realistic setting, i.e., when the sound of the moving finger might interfere with the tracking. Three types of straight touch movements which are respectively parallel to the X-axis, the Y-axis, and at 45° with both axes were evaluated. Out of the above 25 touch points, we selected 12 touch positions which formed 4 isosceles right triangles, whose edges were leveraged as the continuous touch trajectory (Figure 5B). Particularly, three edges of each triangle respectively belonged to one type of straight touch movement. For all triangles, the gestures are always performed clockwise, which allowed us to test the gestures, back and forth, for all directions.

5.2 Procedure

We sequentially conducted the study on discrete touch localisation, continuous touch tracking and touch localisation on different materials. Since there is no gestures performed or learned, we assumed that there was no carry-over effect. For discrete touch localisation, participants were asked to touch each of the testing position first, then we successively read 100 data points. For continuous touch localisation, participants were asked to ensure that their finger was not touching the surface at the start. After the system started to

⁷<https://www.grc.nasa.gov/www/k-12/airplane/sound.html>

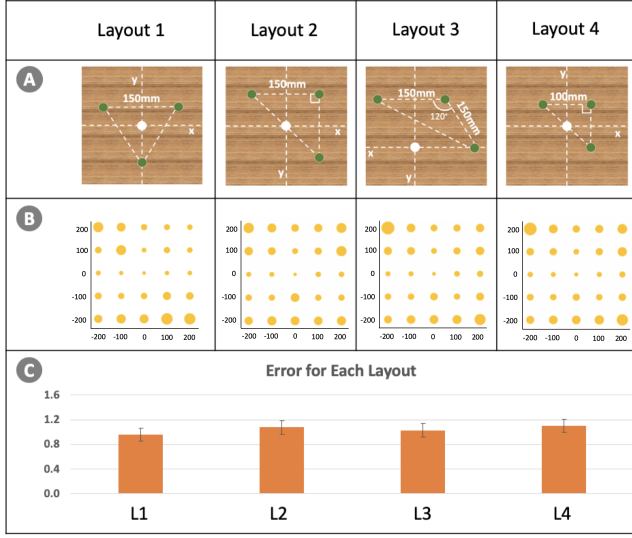


Figure 6: 25 touch points were tested under four different system layouts. A,B: visualisation of average discrete touch error for these different layouts; C: average localisation error with standard deviation with respect to touch points and users respectively. The highlighted area provides a high touch localisation accuracy.

read the data, participants were then required to touch the table and perform the corresponding moving gestures. After finishing the gesture, they needed to lift their finger again. Then the data recording would be stopped by the researcher. Touch gesture along one edge of a triangle was regarded as one single gesture. Three gestures in the same triangle were tested successively, always in a clockwise sequence. The order of testing 4 triangles was randomised. All gestures were repeated twice for each participant. For surface effect evaluation, we conducted the same procedure as layout evaluation. Overall, each participant had 25 (locations) $\times$ 4 (layouts) + 25 (locations) $\times$ 3 (materials) = 175 trials for discrete touch and 12 (locations) $\times$ 2 (repetitions) = 24 trials for continuous touch.

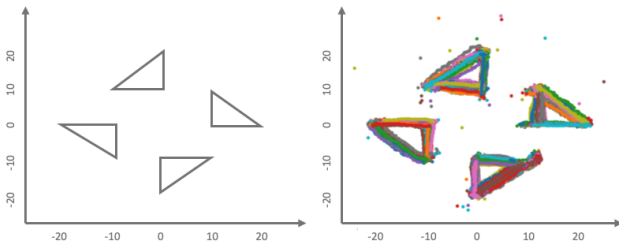


Figure 7: Overall continuous touch data for all the users. Left: reference shapes. Right: reported touch locations colour-coded by user.

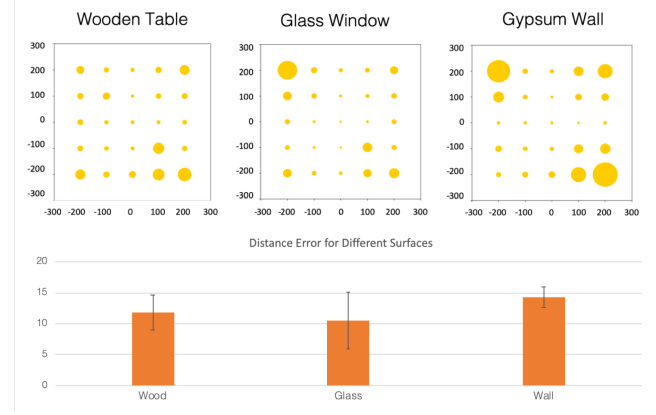


Figure 8: Touch localisation accuracy was evaluated on 3 different everyday surfaces: wooden table, glass window and interior wall made from gypsum board.

5.3 Results

5.3.1 Different Layouts: Figure 6 shows the discrete touch accuracy, as well as the standard deviation, for all 25 touch points for all the layouts. The average errors are 9.62mm ($SD = 1.69mm$) for layout 1, 10.76mm ($SD = 1.77mm$) for layout 2, 10.31mm ($SD = 1.74mm$) for layout 3, 11.03mm ($SD = 1.72mm$) for layout 4 and 10.43mm for all the layouts. Based on a one-way ANOVA analysis, there was no main effect of layouts on discrete touch accuracy ($F(3, 966) = 2.58, p > 0.05$). According to Figure 6A, large prediction error mainly happens to touch points at the corners, indicating that our system starts to introduce a relatively high error when the touch points are around 280mm away from the origin. Particularly, the up-left touch point is subject to the largest average error, which is 17.88 mm. Considering that the contact area of our finger with the table is around 10mm, we think this error is still acceptable. Thus, we conclude that this 400mm by 400mm square area can provide an acceptable accuracy for interaction.

5.3.2 Continuous Touch Movement: Figure 7 shows the overall continuous touch contours for all the users. It is clear that, in general, system was able to capture the continuous touch gestures. There are some touch points exhibiting a large deviation from the intended trajectory. We found that most of this points are 3cm away from the segment. In total, these contribute to <1.08% of overall touch points. Also, the mean distance between two successive touch points was 2.1mm ($SD = 0.1mm$) at a normal moving speed.

5.3.3 Different Materials: Figure 8 shows the discrete touch accuracy, as well as the standard deviation, for all 25 touch points for all the surfaces. The average errors are 11.80mm ($SD = 2.82mm$) for natural wood table, 10.49mm ($SD = 4.61mm$) for glass window and 14.25mm ($SD = 1.70mm$) for interior gypsum wall. Similar with our previous study, large prediction error mainly happens to touch points at the corners, where the touch points are around 280mm away from the origin. The accuracy indicates that our system has the potential to be used in real world.

6 STUDY 2: FORCE SENSING

The aim of this study was to validate prediction of the pseudo force level of the users' touch based on signal strength on different materials. We first collected the force data through a force-sensitive resistor (FSR) as the ground truth, and the signal strength sensed by our VersaTouch system. Then we modelled the relationship between the pseudo force and the signal strength and location of touch, with a mixture of empirical (for the relationship between signal and force at the touch location) and physics-based (for attenuation of the signal from the touch location to the sensors) models. We fitted the model to the collected data and evaluated its accuracy.

6.1 Apparatus

We collected data on three surfaces (wooden table, glass window and gypsum interior wall), at four discrete distances from the sensor (100mm, 200mm and 300mm). We used a force sensing resistor (FSR) (FSR406⁸) to capture the ground truth of the touch force. The logarithm of the resistance value of the FSR is linearly associated with the logarithm of the force, which we used as the pseudo force. During the study, the FSR was placed between the finger and the surface. We used the single-sensor mode of VersaTouch in this study to keep the apparatus simple, since it is already capable of sensing the distance of the touch location to the sensor.

6.2 Procedure

Ten participants were recruited for the study. Each participant was required to perform touch actions of different force levels at 4 different points, which were respectively 100mm, 200mm and 300mm away from the sensor, and were marked on the surfaces. For each surface, participants needed to exert the touch force from lightest to heaviest on each point and back to lightest until the finger lifted from the surface and VersaTouch detected a finger release. An FSR was firmly attached to each touch point, and the pseudo force was visualised to the user while they performed the action, so as to ensure the full coverage of the force range they could output. We collected all the data when the pseudo force was between 0 and 1, which correspond to the force level between 60g and 1500g according to the datasheet of the FSR⁹. 60g is the normal force when people perform a touch action [29]. 1500g is the force level that higher than most industrial tactile sensors [43]. For each touch point, we collected 500 data points both from the FSR as ground truth and VersaTouch (signal strength and measured distance).

6.3 Pseudo Force Modelling and Prediction

Our modelling of the pseudo force takes two steps. First, we empirically determine the relationship between the signal strength and the pseudo force at fixed distances. Second, we introduce distance as a variable into the model, by applying the physical model of attenuation of mechanical vibration. Finally, we are able to predict pseudo force as a function of observed signal strength at the sensor and the touch distance measured by VersaTouch.

6.3.1 Signal Strength and Pseudo Force at Fixed Distances: Figure 9 plots the signal strength against the pseudo force (logarithm of resistance value) measured by the FSR. The pseudo force and signal

readings of VersaTouch were well fitted with polynomial (order 2 and order 3) and logarithmic regression models. However, polynomial functions are not monotonic, resulting that two different sensor readings might map to a same force level. Thus, a logarithmic function is a better model for force prediction. Since different fingers have distinctive influences on the signal intensity, we trained model for each participant. We used 80% of the data to train the model and other 20% to test the accuracy. The average prediction error was shown in Figure 9b. When the distance was no more than 200mm, the error of pseudo force was less than 0.075. When the distance reached 300mm, the error was around 0.08. As expected, the sensing error increased with the distance. However, two-way ANOVA analysis showed that there was no significant difference either among surfaces ($F(2, 81) = 2.54, p > 0.05$) or distances ($F(2, 81) = 1.65, p > 0.05$). To measure how well the data fits the model, we also calculated the root mean square error (RMSE) and the coefficient of determination (r^2) between the estimated force with the pseudo force. The r^2 values ranged from 0.86 to 0.97 and RMSE ranged from 0.05 to 0.08, which was consistent with the above observation.

6.3.2 Adding Distance to the Model: Assuming that a logarithmic model can predict the pseudo force at a single touch point, we then explored a model to predict force at different distances from the sensor. First, at the touch location, the relation between the signal and pseudo force is:

$$\ln R = a \ln A_0 + b$$

where R is the FSR value, A_0 is the signal amplitude at the source (i.e., the touch location), a and b are model parameters. When sound travels through a medium, the intensity diminishes with distance. The intensity attenuates based on the following formula¹⁰:

$$A = A_0 e^{-\alpha z}$$

where A is the captured signal amplitude, z is the distance to the sensor. Combined with our logarithmic relationship between signal and pseudo force, the final model to predict pseudo force with captured signal amplitude and distance could be established as:

$$\ln R = a \ln \left(\frac{A}{e^{-\alpha z}} \right) + b$$

which could be rearranged to

$$\ln R = a (\ln A + \alpha z) + b$$

As A and z are our observed values, we can train a multiple linear regression model, with $\ln A$ and z as independent variables to predict the pseudo force $\ln R$. We trained a model for each participant and each material with 80% of the data. Then we tested the sensing accuracy with the remaining 20% data. The result is shown in Figure 9c. Specifically, the accuracy of force sensing with respect to distance was subject to an average error of 0.094 for wooden table, 0.064 for glass window and 0.086 for gypsum wall. One-way ANOVA analysis showed that there was a significant effect of surface. The r^2 ranged from 0.76 to 0.93 for wooden table, 0.91 to 0.95 for glass window, and 0.63 to 0.95 for gypsum wall. The average RMSEs were 0.16, 0.08 and 0.12 respectively for wooden table, glass window and gypsum wall.

⁸<https://www.sparkfun.com/datasheets/Sensors/Pressure/fsrguide.pdf>

⁹<https://www.sparkfun.com/datasheets/Sensors/Pressure/fsrguide.pdf>

¹⁰<https://www.nde-ed.org/EducationResources/CommunityCollege/Ultrasonics/Physics/attenuation.htm>

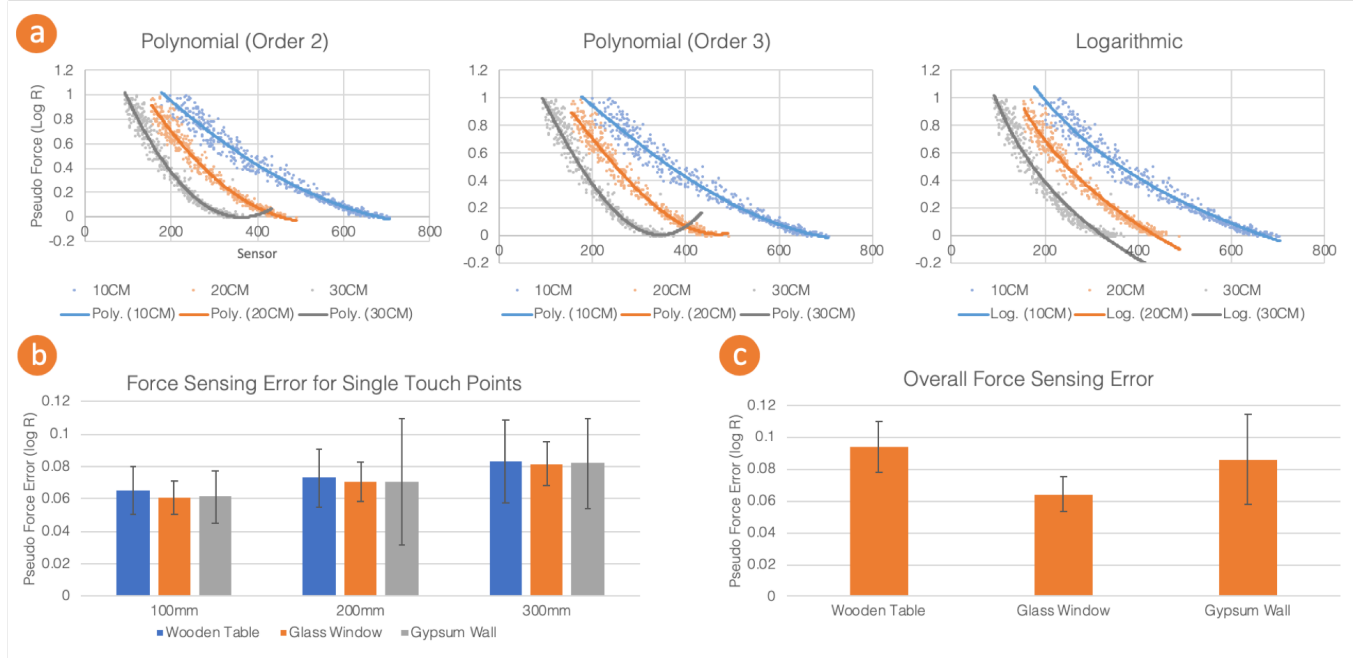


Figure 9: Typical force plot, fitted by different equations and force sensing accuracy analysis result.

7 INTERACTION DESIGN SPACE

Physical Layout. The layout of a VersaTouch system is defined by two variables: the number of sensors and the shape that the sensors form. The number of sensors determines the degrees of freedom for the touch location tracking. With a *single sensor*, 1D distance of the touch point to the sensor could be determined. With *two sensors*, 2D touch location could be tracked, subject to symmetry about the line connecting the two sensors. With *three sensors*, an unambiguous 2D touch location could be determined. *Geometric parameter* of multiple sensors, such as the distance between two sensors or shape of triangle three sensors forms, could also be used to design interactions. The tangibility of physical sensors improves interaction experience in two aspects. First, they are visible to the users, informing the mode of operation. Second, they can be made reactive only to gestures performed around or near them to avoid the Midas touch problem in an always-on interactive system.

Input Modality. VersaTouch supports three modalities of touch input. First, the first-order status of the touch, i.e., *pressed* or *released* could be determined. In contrast, most passive acoustic-based methods could only detect the instantaneous action of finger knock (cf. soft press), which is the second-order transition of touch state from released to pressed, but not the sustained touch status and the release event. Second, the *touch location* of the finger is tracked (with 1 or 2 degrees of freedom depending on the number of sensors explained previously). Third, a continuous *pseudo force* level is tracked regardless of the number of sensors.

Number of Actuators. The active acoustic sensing mechanism of VersaTouch supports *multiple actuators* by time-multiplexing their respective sensing cycles. These multiple actuators could be used for multi-touch tracking, if mounted on different fingers of a

same user, or for user identification, if mounted on different users. Our implementation supports a maximum of 120 sensing cycles per second, which, for example, could support 4 actuators each at a sensing frequency of 30Hz.

Feedback. Multiple types of feedback is available in the VersaTouch system. With a fingernail-mounted actuator, users' *natural tactile perception* of the underlying surface is preserved as the inherent type of feedback of the VersaTouch system. Our choice of bone conductor as the actuator allows for presentation of both *vibrotactile feedback* outside the sensing cycles. Both types of feedback are vibrations in nature, with different frequency ranges to be perceptible by the human ears (20 Hz ~ 20 kHz) or mechanoreceptors (50Hz ~ 300 Hz). Perceptually, they are not perfectly separable, e.g., a vibrotactile stimulus can be heard as a weak bass sound, but this could be an interesting factor for the design of vibro-auditory multimodal feedback. Another design consideration around this vibro-auditory duality is that vibrotactile stimuli could be assigned for more private feedback, and auditory stimuli could be assigned for more public and shared feedback, since the sound could potentially be transmitted and amplified by the underlying surface.

8 INTERACTION TECHNIQUES

To illustrate the interaction potential of VersaTouch, we introduce some interaction techniques built from our design space. Some of these techniques provide novel user experiences, while others improve upon existing techniques. The factors relevant to the design space are italicised.

Shape-coded Widgets. Specific *sensor layouts* in a VersaTouch system could trigger specific functionalities. This is particularly useful if the sensor layout is restricted or defined by the geometry



Figure 10: Different interaction techniques VersaTouch supported, as well as the example applications.

of a physical prop, so as to create hybrid tangible-digital apps. For example, a user can place sensors on the three corners of a physical desk calendar (Figure 10a). Upon detecting the match between the sensor layout and the calendar corners, a calendar app is launched, which allows a user to touch a specific date, through *touch localisation*, to show the agenda on that day on a screen or send it to a smartphone or smartwatch.

1D Dynamic Slider. Two sensors (*2-sensor mode*) could be *tangible markers* to allow 1D sliding input on a segment connecting the two. Users can adjust the control-display ratio of the slider by adjusting the distance between the pair of sensors (Figure 10b), where “control” is the length of physical movement (i.e., in millimetres), and “display” is the change of the logical slider value respective to its whole range. For example, the larger the distance between the two sensors, the longer movement it requires to achieve a same change in the logical slider value. This selection is conveyed to the user via *vibrotactile feedback*.

Toggle Switch With Continuous Parameter. A single sensor itself could also be used as a multi-functional controller. Users can carry one sensor (*1-sensor mode*) and stick it on a wall-light to create a temporary interactive area around the light (Figure 10c), which could be used as a virtual control panel. For example, light could be toggled with quick taps (*input modality*). The colour of the light can be controlled by the distance between the finger and the sensor (*1D localisation*). Force of the touch controls the brightness of the light (*force sensing*). The user also receives *vibrotactile feedback* as a confirmation of selection.

Multi-touch Surface. With *multiple actuators* mounted on different fingers, VersaTouch allows user to perform multi-touch interactions. For example, a user can draw a piano keyboard and register its position on a table by touching specific locations within the drawing (Figure 10d). Users can then interact with the keyboard by touching respective keys (*touch localisation*). Chords can be played with the multi-touch support of the system (*multiple actuators*). Users can perform a light or hard tap *pseudo force* on a key to control the volume of the key stroke.

Identity Awareness. VersaTouch can identify different users (*multiple actuators*), which allows them to perform touch interactions simultaneously and independently of each other. For example, two users could simultaneously use the same VersaTouch system on a same desk as their own touchpad to play multi-player games, such as Whack-a-mole (Figure 10e). The system can distinguish touch interaction from different users and count score for them.

9 LIMITATION AND FUTURE WORK

One obvious limitation of VersaTouch is that the signal we used is of an audible frequency. The intuitive solution would be shift the signal frequency to inaudible range. However, with a lower inaudible frequency, the wavelength of the signal is longer and it requires a much longer time to collect the data in each sensing cycle, resulting in a significant decrease in the touch update speed. Note, that for each iteration, the waiting time is still needed to avoid interference from echoing. Thus, the potential solution is to use signal with higher frequency. In this paper, we evaluate the theory of this method as well as interaction possibilities. To use higher frequency, better hardware, such as faster microcontroller and more sensitive contact microphone, are needed.

Another limitation is that, in our prototype, the data is sent through physical wires, causing different parts of our system to physically connect with each other. Thus, our system is more cumbersome than it needs to be. Future work is required to enable wireless communication, making the system easier to carry around. In fact, with WLAN technology, the delay of data transmission could be realised within 1ms. Other solutions, such as using synchronised clocks on each sensor to compensate the time difference, could also be adopted to solve this problem.

We also found that the environment noise could sometimes affect the performance of the system, especially for large continuous surfaces, such as walls. If an appliance is placed against the wall or construction is being done to the wall, the acoustic signals generated by them could potentially affect the performance.

In the future, further research needs to be conducted on the signal decomposition to better distinguish environmental noise from the use signal. Moreover, the system integration could be enhanced, so the device is minimised to a level at that it is almost invisible to the user, becoming an integral extension of the own body[6]. Then, we could enable a reflexive interaction [21] in our everyday day life’s, such as silencing a vibrating phone while drawing a quick rejecting swipe gesture on any surface.

10 CONCLUSION

In this paper, we introduced VersaTouch, a portable and versatile system that enables various touch interactions on everyday surfaces. VersaTouch is versatile in terms of sensing ability, system layout, interaction possibilities, feasibility for everyday surfaces, and provision of vibrotactile feedback. Benchmarking results showed that our system is able to offer these interaction modalities with a relatively high accuracy. Also, we built four VersaTouch-enabled fully functional demos to show the interaction possibilities of our system.

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